H2 Mathematics · Set 1 · Paper 2 · A-Level format · Worked solutions

100 marks

1. A geometric series is given by $1+\frac{x-1}{x+2}+\left(\frac{x-1}{x+2}\right)^2+\left(\frac{x-1}{x+2}\right)^3+\cdots$, where x is real and $x \neq -2$.

   (a) Find the set of values of x for which the series converges. [2]

   (b) Find the sum to infinity of the series, giving your answer as a single fraction in terms of x. [1]

   (c) Find exactly the set of values of x for which the sum to infinity of the series exceeds $\frac{2}{x}$. [4]

Solution:

(a) The series is geometric with first term 1 and common ratio $r=\frac{x-1}{x+2}$, so it converges if and only if $|r|<1$.

$\left|\frac{x-1}{x+2}\right|<1 \Leftrightarrow |x-1|<|x+2|$ (as $x \neq -2$).

Both sides are non-negative, so squaring keeps the inequality: $(x-1)^2<(x+2)^2 \Leftrightarrow x^2-2x+1<x^2+4x+4 \Leftrightarrow -6x<3 \Leftrightarrow x>-\frac{1}{2}$.

(Equivalently: x is nearer to 1 than to -2 on the number line, so it lies to the right of their midpoint $-\frac{1}{2}$.)

The series converges for $\left\{x \in \mathbb{R}: x>-\frac{1}{2}\right\}$. [M1 for $|x-1|<|x+2|$ or $-1<\frac{x-1}{x+2}<1$; A1]

(b) Sum to infinity $=\frac{1}{1-r}=\frac{1}{1-\frac{x-1}{x+2}}=\frac{x+2}{(x+2)-(x-1)}=\frac{x+2}{3}$. [B1]

(c) The decision: a sum to infinity exists only where the series converges, that is for $x > -\frac{1}{2}$ by part (a). The expression $\frac{x+2}{3}$ is defined for every x, but it is the sum of the series only on that set, so the inequality is solved on $x > -\frac{1}{2}$ and not on all of $\mathbb{R}$.

$\frac{x+2}{3} > \frac{2}{x} \Leftrightarrow \frac{x+2}{3} - \frac{2}{x} > 0 \Leftrightarrow \frac{x^2+2x-6}{3x} > 0$. [M1: one side, single fraction, not cross-multiplied]

$x^2 + 2x - 6 = 0 \Leftrightarrow x = -1 \pm \sqrt{7}$, so the inequality is $\frac{(x+1+\sqrt{7})(x+1-\sqrt{7})}{3x} > 0$, with critical values $-1 - \sqrt{7}$ (≈ -3.65), 0 and $\sqrt{7} - 1$ (≈ 1.65).

Sign test: for $x < -1 - \sqrt{7}$ the fraction is negative; for $-1 - \sqrt{7} < x < 0$ it is positive; for $0 < x < \sqrt{7} - 1$ it is negative; for $x > \sqrt{7} - 1$ it is positive.

So $\frac{x+2}{3} > \frac{2}{x} \Leftrightarrow -1 - \sqrt{7} < x < 0$ or $x > \sqrt{7} - 1$. [A1: the raw intervals from the sign test]

Restrict to $x > -\frac{1}{2}$: since $-1 - \sqrt{7} < -\frac{1}{2} < 0$, the first interval is cut to $-\frac{1}{2} < x < 0$; the second is unchanged because $\sqrt{7} - 1 > -\frac{1}{2}$. [M1: the inequality restricted to the convergence set $x > -\frac{1}{2}$, where alone $\frac{x+2}{3}$ is the sum]

Required set: $\left\{x \in \mathbb{R}: -\frac{1}{2} < x < 0 \text{ or } x > \sqrt{7} - 1\right\}$. [A1: the cut set. A candidate who solves $\frac{x+2}{3} > \frac{2}{x}$ on all of $\mathbb{R}$ scores M1 A1 M0 A0, 2 of the 4 marks]

The values $-1 - \sqrt{7} < x \leq -\frac{1}{2}$ satisfy $\frac{x+2}{3} > \frac{2}{x}$ but are rejected: there $|r| \geq 1$ (for example $x = -1$ gives $r = -2$), so the series has no sum to infinity; that interval even contains $x = -2$, where the series is not defined.

Alternative (working on the convergence set from the start): for $-\frac{1}{2} < x < 0$ the sum $\frac{x+2}{3}$ lies between $\frac{1}{2}$ and $\frac{2}{3}$, so it is positive while $\frac{2}{x} < 0$, and every such x satisfies the inequality; for $x > 0$, multiplying by $3x > 0$ gives $x^2 + 2x - 6 > 0$, so $x > \sqrt{7} - 1$ (the other root, $-1 - \sqrt{7}$, is negative). The same set results.

Common errors: solving on all of $\mathbb{R}$ gives $-1 - \sqrt{7} < x < 0$ or $x > \sqrt{7} - 1$ (convergence forgotten, final A0); cross-multiplying by x as though it were positive gives $x^2 + 2x - 6 > 0$ and loses the interval $-\frac{1}{2} < x < 0$.

✓ **Check:** $x = -0.4$: $r = -0.875$, sum $= 0.533 > \frac{2}{-0.4} = -5$ (in the set); $x = 1.6$: sum $= 1.2 < 1.25$ (not in the set); $x = 1.7$: sum $= 1.233 > 1.176$ (in the set); $x = -1$: $\frac{x+2}{3} = \frac{1}{3} > -2$ but $r = -2$, no sum (rejected).

2. The curve C has equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = \frac{\ln x}{x}$ for $x > 0$.

(a) Find the exact coordinates of the stationary point of C, and show that it is a maximum point. [3]

(b) Determine, with justification, which of 3^π and π^3 is greater. [3]

(c) Show that $x^{\mathrm{e}} \leq \mathrm{e}^x$ for all $x > 0$. [1]

(d) Find, with justification, all the positive integers n for which $n^{\frac{1}{n}} \geq \sqrt{2}$. [2]

Solution:

(a) By the quotient rule, $f'(x) = \frac{x \cdot \frac{1}{x} - \ln x}{x^2} = \frac{1 - \ln x}{x^2}$.

At a stationary point $f'(x) = 0$, so $\ln x = 1$, giving $x = e$ (the only solution), and $y = \frac{\ln e}{e} = \frac{1}{e}$. The stationary point is $\left(e, \frac{1}{e}\right)$.

Nature: $x^2 > 0$, so $f'(x)$ has the sign of $1 - \ln x$. For $0 < x < e$, $\ln x < 1$ and $f'(x) > 0$; for $x > e$, $\ln x > 1$ and $f'(x) < 0$. f' changes from positive to negative at $x = e$, so the point is a maximum. (Or: $f''(x) = \frac{2\ln x - 3}{x^3}$, and $f''(e) = -\frac{1}{e^3} < 0$, so a maximum.)

[M1 quotient rule to $\frac{1 - \ln x}{x^2}$; A1 $\left(e, \frac{1}{e}\right)$ exact; A1 maximum by a correct first- or second-derivative test.]

(b) Since ln is increasing, $3^\pi > \pi^3 \Leftrightarrow \pi \ln 3 > 3 \ln \pi \Leftrightarrow \frac{\ln 3}{3} > \frac{\ln \pi}{\pi}$ (dividing by $3\pi > 0$) $\Leftrightarrow f(3) > f(\pi)$.

From (a), $f'(x) = \frac{1 - \ln x}{x^2} < 0$ for $x > e$, so f is strictly decreasing for $x \geq e$. Since $e < 3 < \pi$, $f(3) > f(\pi)$.

Hence $3^\pi > \pi^3$.

[M1 the comparison of powers turned into $f(3)$ against $f(\pi)$ unprompted; M1 f decreasing for $x > e$ with $e < 3 < \pi$; A1 $3^\pi > \pi^3$. Calculator values ($3^\pi \approx 31.5$, $\pi^3 \approx 31.0$) alone earn no marks. The maximum alone does not decide it (both 3 and π lie beyond e).]

(c) From (a), f increases for $0 < x < e$ and decreases for $x > e$, so $\frac{1}{e}$ is the greatest value of f: $\frac{\ln x}{x} \leq \frac{1}{e}$ for all $x > 0$.

Multiplying by ex, which is positive: $e \ln x \leq x$, that is, $\ln(x^e) \leq x = \ln(e^x)$.

Since ln is an increasing function, $x^e \leq e^x$ for all $x > 0$ (with equality only at $x = e$).

[B1 $\frac{\ln x}{x} \leq \frac{1}{e}$ used as the greatest value of f for ALL $x > 0$ and rearranged through $e \ln x \leq x$ to the printed result, the multiplier ex noted positive.]

(d) Since ln is increasing and $n > 0$, $n^{\frac{1}{n}} \geq \sqrt{2} = 2^{\frac{1}{2}} \Leftrightarrow \frac{1}{n} \ln n \geq \frac{1}{2} \ln 2 \Leftrightarrow f(n) \geq f(2)$.

Also $\sqrt{2} = (2^2)^{\frac{1}{4}} = 4^{\frac{1}{4}}$, so $f(4) = \frac{\ln 4}{4} = \frac{2\ln 2}{4} = \frac{\ln 2}{2} = f(2)$ exactly.

$n = 1$: $f(1) = 0 < f(2)$, so $n = 1$ fails. $n = 2$: equality holds, so $n = 2$ satisfies the inequality.

For $n \geq 3$ we have $n > e$, where f is strictly decreasing. $n = 3$: $3 < 4$ gives $f(3) > f(4) = f(2)$, so $n = 3$ satisfies it. $n = 4$: equality, so $n = 4$ satisfies it. $n \geq 5$: $n > 4$ gives $f(n) < f(4) = f(2)$, so every $n \geq 5$ fails.

Therefore $n = 2,3,4$ only.

[B1 the inequality as $f(n) \geq f(2)$ with $f(4) = f(2)$ seen exactly (from $\sqrt{2} = 4^{\frac{1}{4}}$); A1 $n = 2,3,4$ with every $n \geq 5$ excluded because f decreases for $x > e$. A table of calculator values alone cannot settle all $n \geq 5$, nor the exact tie at $n = 4$ (1.41421 ... for both $n = 2$ and $n = 4$), and scores at most 1. A candidate who gives the tie $f(2) = f(4)$ without excluding every $n \geq 5$ earns 1 of the 2 marks.]

3. The function f is defined by $\mathrm{f}: x \mapsto 1 - x^2,\ x \in \mathbb{R},\ x \geq 0$.

(a) Find $\mathrm{f}^{-1}(x)$ and state the domain of f^{-1}. [2]

(b) Sketch, on the same diagram, the graphs of $y = \mathrm{f}(x)$ and $y = \mathrm{f}^{-1}(x)$, showing clearly the relationship between the two graphs and stating the coordinates of the points where each graph meets the axes. [2]

(c) Find the exact coordinates of the points at which the graphs of $y = \mathrm{f}(x)$ and $y = \mathrm{f}^{-1}(x)$ meet, and show that there are no other such points. [4]

(d) The function g is increasing for all values of x in its domain, so that g^{-1} exists. Explain why every point at which the graphs of $y = \mathrm{g}(x)$ and $y = \mathrm{g}^{-1}(x)$ meet lies on the line $y = x$. [2]

(e) The function h is defined by $\mathrm{h}: x \mapsto 1 - x^2,\ x \in \mathbb{R},\ x \leq 0$. Use the result of part (d) to find the exact coordinates of the point at which the graphs of $y = \mathrm{h}(x)$ and $y = \mathrm{h}^{-1}(x)$ meet. [1]

Solution:

(a) Let $y = 1 - x^2$, $x \geq 0$. Then $x^2 = 1 - y$, and $x = \sqrt{1 - y}$, taking the positive square root because $x \geq 0$.

So $\mathrm{f}^{-1}(x) = \sqrt{1 - x}$.

f is decreasing for $x \geq 0$, with $\mathrm{f}(0) = 1$ and $\mathrm{f}(x) \to -\infty$ as $x \to \infty$, so $R_\mathrm{f} = (-\infty, 1]$.

$D_{\mathrm{f}^{-1}} = R_\mathrm{f} = (-\infty, 1]$, i.e. $x \leq 1$.

[B1 $\sqrt{1 - x}$, the positive root; B1 the domain.]

(b) One diagram, equal scales, the line $y = x$ drawn (dashed) as the mirror line.

$y = \mathrm{f}(x)$: the right half of the parabola $y = 1 - x^2$, starting at the end-point $(0,1)$ on the y-axis (horizontal tangent there), decreasing and concave downwards, crossing the x-axis at $(1,0)$ and continuing downwards as x increases.

$y = \mathrm{f}^{-1}(x) = \sqrt{1 - x}$: the reflection of $y = \mathrm{f}(x)$ in the line $y = x$, starting at the end-point $(1,0)$ on the x-axis (vertical tangent there), crossing the y-axis at $(0,1)$ and rising as $x \to -\infty$.

Each graph meets the axes at $(0,1)$ and $(1,0)$, so the two graphs share both of these points; between $x = 0$ and $x = 1$ the graph of f is first above that of f^{-1} and then below it, crossing it on the line $y = x$.

[B1 each curve with its end-point and its intercepts; B1 the two graphs clearly reflections of each other in $y = x$.]

(c) A point (p, q) lies on both graphs when $q = \mathrm{f}(p)$ and $q = \mathrm{f}^{-1}(p)$, with p in both domains: $p \geq 0$ and $p \leq 1$.

Solving $\mathrm{f}(x) = x$ finds only the meeting points ON the line $y = x$. Here f is decreasing, and the sketch in (b) already shows two common points off that line, so the equation to solve is $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$ itself:

$1 - x^2 = \sqrt{1 - x}$, $0 \leq x \leq 1$.

For $0 \leq x \leq 1$ both sides are non-negative, so squaring gives an equivalent equation:

$(1 - x^2)^2 = 1 - x$

$(1 - x)^2(1 + x)^2 - (1 - x) = 0$

$(1 - x)[(1 - x)(1 + x)^2 - 1] = 0$

$(1 - x)(1 + x - x^2 - x^3 - 1) = 0$

$(1 - x)(x - x^2 - x^3) = 0$

$x(1 - x)(1 - x - x^2) = 0$

[Alternatively: (p, q) is on the graph of f^{-1} exactly when $\mathrm{f}(q) = p$, so a common point has $\mathrm{f}(\mathrm{f}(p)) = p$: $1 - (1 - x^2)^2 = x$, i.e. $x^4 - 2x^2 + x = 0$, $x(x - 1)(x^2 + x - 1) = 0$ — the same equation. The sketch shows that $x = 0$ and $x = 1$ are roots, which gives the factor $x(x - 1)$ at once.]

So $x = 0$, $x = 1$, or $x^2 + x - 1 = 0$, giving $x = \frac{-1 \pm \sqrt{5}}{2}$.

$x = \frac{-1 - \sqrt{5}}{2} \approx -1.62$ is not in $0 \leq x \leq 1$ (it is outside the domain of f): rejected. [There $1 - x^2 < 0 < \sqrt{1 - x}$, so it came only from squaring.]

$x = \frac{\sqrt{5} - 1}{2} \approx 0.618$ lies in $0 \leq x \leq 1$.

y-coordinates: $\mathrm{f}(0) = 1$; $\mathrm{f}(1) = 0$; for $x = \frac{\sqrt{5} - 1}{2}$, $x^2 = 1 - x$, so $\mathrm{f}(x) = 1 - x^2 = x = \frac{\sqrt{5} - 1}{2}$.

The quartic has only these four roots and one of them is rejected, so the graphs meet at exactly three points: $(0,1)$, $(1,0)$ and $\left(\frac{\sqrt{5} - 1}{2}, \frac{\sqrt{5} - 1}{2}\right)$.

Two of them lie off the line $y = x$: $\mathrm{f}(0) = 1$ and $\mathrm{f}(1) = 0$, so $(0,1)$ and $(1,0)$ are both on the graph of f, and as each is the reflection of the other in $y = x$, both are also on the graph of f^{-1}.

[M1 equating $\mathrm{f}(x)$ and $\mathrm{f}^{-1}(x)$, or $\mathrm{f}^2(x) = x$ — NOT $\mathrm{f}(x) = x$; M1 squaring and factorising the quartic; A1 the four roots with $\frac{-1 - \sqrt{5}}{2}$ rejected for a stated reason; A1 all three points exact. A candidate who solves only $\mathrm{f}(x) = x$ earns at most 1 mark.]

(d) Let (a, b) be a point on both graphs. Then $b = \mathrm{g}(a)$, and $b = \mathrm{g}^{-1}(a)$ gives $\mathrm{g}(b) = a$.

Suppose $a < b$. Since g is increasing, $\mathrm{g}(a) < \mathrm{g}(b)$, that is $b < a$ — a contradiction.

Suppose $a > b$. Then $\mathrm{g}(a) > \mathrm{g}(b)$, that is $b > a$ — a contradiction.

Hence $a = b$, and the point (a, b) lies on the line $y = x$.

[For the decreasing f of part (c), $a < b$ gives $\mathrm{f}(a) > \mathrm{f}(b)$, that is $b > a$, which is no contradiction — which is why a point such as $(0,1)$, with $a = 0$ and $b = 1$, can occur.]

[B1 $\mathrm{g}(a) = b$ and $\mathrm{g}(b) = a$; B1 the contradiction from g increasing, in both cases or 'without loss of generality'.]

(e) For $x_1 < x_2 \leq 0$, $x_1^2 > x_2^2$, so $\mathrm{h}(x_1) < \mathrm{h}(x_2)$: h is increasing. By part (d), the meeting point lies on $y = x$, so $1 - x^2 = x$ with $x \leq 0$:

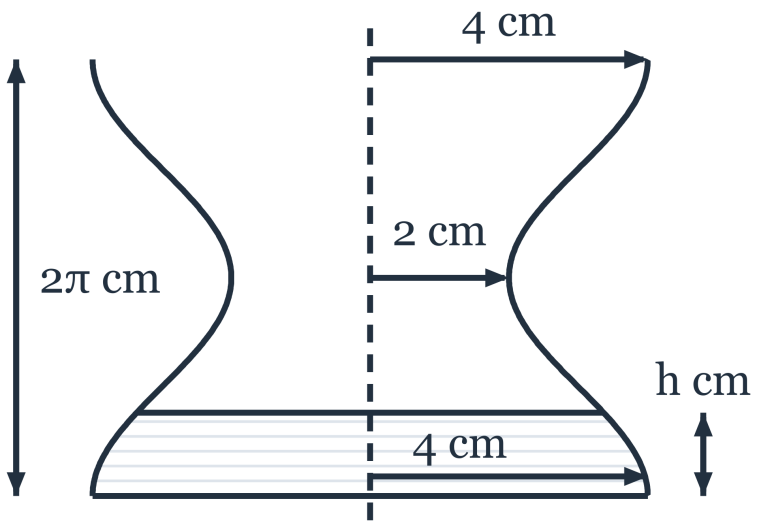
$x^2 + x - 1 = 0$, $x = \frac{-1 \pm \sqrt{5}}{2}$, and $x \leq 0$ gives $x = \frac{-1-\sqrt{5}}{2}$ (also in the domain $x \leq 1$ of h^{-1}).

Point: $\left(-\frac{1+\sqrt{5}}{2}, -\frac{1+\sqrt{5}}{2}\right)$ — the root rejected in part (c) is the meeting point for the other branch of $y = 1 - x^2$.

[Check: $\mathrm{h}^{-1}(x) = -\sqrt{1-x}$; at $x = -\frac{1+\sqrt{5}}{2}$, $1 - x = \frac{3+\sqrt{5}}{2} = \left(\frac{1+\sqrt{5}}{2}\right)^2$, so $\mathrm{h}^{-1}(x) = -\frac{1+\sqrt{5}}{2} = x$.]

[B1.]

4. The inside of a vase has a flat horizontal base and an open top at a height of 2π cm above the base. At height y cm above the base, where $0 \leq y \leq 2\pi$, the horizontal cross-section of the inside of the vase is a circle of radius $(3 + \cos y)$ cm (see diagram). The vase is initially empty and water is poured into it at a constant rate of 12π cm^3 per second; at time t seconds after pouring begins, the depth of the water is h cm.



(a) Find the rate at which the depth of the water is increasing at the instant when $h = \frac{1}{2}\pi$. Find also the rate of change of the radius of the water surface at this instant. [5]

(b) Find the exact time taken to fill the vase. [4]

(c) Find the exact depth of the water at the instant when half of the time found in part (b) has passed. Justify your answer. [2]

(d) Without further differentiation, find the greatest and least values of the rate at which the depth of the water increases for $0 \leq h \leq 2\pi$, and state the depths of the water at which they occur. [2]

Solution:

(a) Let the radius of the water surface be r cm, so $r = 3 + \cos h$.

No formula for the volume V cm^3 of water is given. The water is made of thin horizontal circular slices: the volume of water is $V = \int_0^h \pi\,(3 + \cos y)^2\,\mathrm{d}y$ (the region between the curve $x = 3 + \cos y$ and the y-axis, for $0 \le y \le h$, rotated about the y-axis). Raising the depth by a small δh adds a thin layer of volume $\delta V \approx \pi(3 + \cos h)^2\,\delta h$, so $\frac{\mathrm{d}V}{\mathrm{d}h} = \pi(3 + \cos h)^2$, the area of the water surface. [B1]

(Or integrate first: $V = \pi \int_0^h \left(9 + 6\cos y + \frac{1}{2} + \frac{1}{2}\cos 2y\right)\mathrm{d}y = \pi\left(\frac{19}{2}h + 6\sin h + \frac{1}{4}\sin 2h\right)$, so $\frac{\mathrm{d}V}{\mathrm{d}h} = \pi\left(\frac{19}{2} + 6\cos h + \frac{1}{2}\cos 2h\right) = \pi(9 + 6\cos h + \cos^2 h) = \pi(3 + \cos h)^2$.)

$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}t} \div \frac{\mathrm{d}V}{\mathrm{d}h} = \frac{12\pi}{\pi(3+\cos h)^2} = \frac{12}{(3+\cos h)^2}$. [M1]

When $h = \frac{1}{2}\pi$, $\cos h = 0$, so $\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{12}{9} = \frac{4}{3}$: the depth is increasing at $\frac{4}{3}$ cm per second. [A1]

$\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{\mathrm{d}r}{\mathrm{d}h} \times \frac{\mathrm{d}h}{\mathrm{d}t} = -\sin h \times \frac{12}{(3+\cos h)^2}$. [M1]

When $h = \frac{1}{2}\pi$, $\sin h = 1$, so $\frac{\mathrm{d}r}{\mathrm{d}t} = -\frac{4}{3}$: the radius of the water surface is DECREASING at $\frac{4}{3}$ cm per second (for $0 < h < \pi$ the vase narrows as the water rises). [A1] (The answer $+\frac{4}{3}$, or 'increasing', earns A0.)

(b) The vase is full when $h = 2\pi$. Its capacity is

$\int_0^{2\pi} \pi\,(3 + \cos y)^2\,\mathrm{d}y = \pi \int_0^{2\pi} (9 + 6\cos y + \cos^2 y)\,\mathrm{d}y$ [M1]

$= \pi \int_0^{2\pi} \left(9 + 6\cos y + \frac{1}{2} + \frac{1}{2}\cos 2y\right)\mathrm{d}y$, using $\cos^2 y = \frac{1}{2}(1 + \cos 2y)$ [M1]

$= \pi\left[\frac{19}{2}y + 6\sin y + \frac{1}{4}\sin 2y\right]_0^{2\pi} = \pi(19\pi + 0 + 0) = 19\pi^2$ cm^3. [A1]

(Over the full period $0 \le y \le 2\pi$ the $\cos y$ term contributes nothing and $\cos^2 y$ contributes π.)

Time taken $= \frac{19\pi^2}{12\pi} = \frac{19\pi}{12}$ seconds (≈ 4.97 s). [A1]

(A calculator value alone, 187.5 cm^3 or 4.97 s, is not exact and earns only the first M1 if the integral is set up.)

(c) Water enters at a constant rate, so when half of $\frac{19\pi}{12}$ s, that is $\frac{19\pi}{24}$ s, has passed, the vase holds half its capacity, $\frac{19\pi^2}{2}$ cm^3. [B1]

The radius at height $2\pi - y$ is $3 + \cos(2\pi - y) = 3 + \cos y$, the radius at height y: the inside of the vase is symmetric about the horizontal plane at height π, its narrowest section (radius 2). So the part of the vase below height π holds exactly half the capacity, and since V increases as h increases, no other depth gives half: the depth is π cm. [B1]

(Check with V from (a): $V = \pi\left(\frac{19}{2}\pi + 6\sin\pi + \frac{1}{4}\sin 2\pi\right) = \frac{19\pi^2}{2}$ when $h = \pi$. A calculator solution of $\frac{19}{2}h + 6\sin h + \frac{1}{4}\sin 2h = \frac{19}{2}\pi$ giving 3.14 earns the second mark only when the exact value π is stated and verified.)

(d) From (a), $\frac{dh}{dt} = \frac{12}{(3+\cos h)^2}$: the depth rises fastest where the water surface is smallest and slowest where it is largest. Since $-1 \leq \cos h \leq 1$, $2 \leq 3 + \cos h \leq 4$.

Greatest rate $= \frac{12}{2^2} = 3$ cm per second, when $\cos h = -1$, that is $h = \pi$ (the narrowest section). [B1]

Least rate $= \frac{12}{4^2} = \frac{3}{4}$ cm per second, when $\cos h = 1$, that is $h = 0$ (as pouring begins) and $h = 2\pi$ (as the vase becomes full). [B1]

(The greatest rate is 4 times the least: $\left(\frac{4}{2}\right)^2$, the ratio of the areas of the widest and narrowest sections. By (c), the depth rises fastest exactly half way through the filling, when $t = \frac{19\pi}{24}$.)

5. The integers 1, 2, 3, 4 and 5 are arranged in a row, each integer being used exactly once. Such an arrangement is called a 'ridge' if the integers to the left of 5 increase from left to right and the integers to the right of 5 decrease from left to right. For example, 2 3 5 4 1 and 5 4 3 2 1 are ridges, but 2 5 3 4 1 is not.

(a) Find the number of ridges in which 5 is in the middle position. [1]

(b) Show that there are 16 ridges. [2]

(c) Explain why, in every ridge, 1 is at one of the two ends. [1]

(d) The integers $1,2,3,\ldots,n$, where $n \geq 2$, are now arranged in a row in a random order, and a ridge is defined in the same way, with n in place of 5. Find the smallest value of n for which the probability that the arrangement is a ridge is less than 0.01. [3]

Solution:

(a) With 5 in the middle position there are two integers on each side of it. Choose which 2 of 1,2,3,4 go to the left of 5: ${}^4C_2 = 6$ ways. The two on the left must be written in increasing order and the two on the right in decreasing order, so each choice gives exactly one ridge.
Number of ridges $= 6$.

(b) In a ridge, each of the integers 1,2,3,4 is either to the left or to the right of 5: $2 \times 2 \times 2 \times 2 = 2^4$ ways of splitting them into the two sides. Once the two sides are decided, the integers on the left must be in increasing order and those on the right in decreasing order, so each split gives exactly one ridge, and different splits give different ridges.
Number of ridges $= 2^4 = 16$ (shown).
[Alternatively, with k integers to the left of 5 ($k = 0,1,2,3,4$) there are ${}^4C_k$ ridges, as in (a), so the total is ${}^4C_0 + {}^4C_1 + {}^4C_2 + {}^4C_3 + {}^4C_4 = 1 + 4 + 6 + 4 + 1 = 16$.]

(c) 1 is not 5, so it is on one side of 5, and it is the smallest of the integers. If 1 is to the left of 5: the integers there increase from left to right, so the smallest of them is first, and 1 is at the left-hand end. If 1 is to the right of 5: the integers there decrease from left to right, so the smallest of them is last, and 1 is at the right-hand end. Either way 1 is at one of the two ends.

(d) As in (b), each of $1,2,\ldots,n-1$ is either to the left or to the right of n, and the order on each side is then forced, so there are 2^{n-1} ridges.

[Alternatively, by (c): removing 1 from a ridge of $1,2,\ldots,n$ leaves a ridge of $2,3,\ldots,n$, and 1 can be placed at either end of any such ridge, so the number of ridges doubles each time n increases by 1; with 1 ridge when $n=1$, there are 2^{n-1}.]

There are $n!$ equally likely arrangements, so

$$\mathrm{P(ridge)} = p_n = \frac{2^{n-1}}{n!}$$

$\frac{p_{n+1}}{p_n} = \frac{2^n}{(n+1)!} \times \frac{n!}{2^{n-1}} = \frac{2}{n+1} < 1$ for $n \geq 2$, so p_n decreases as n increases.

$$p_6 = \frac{32}{720} = 0.0444$$

$$p_7 = \frac{64}{5040} = 0.0127 > 0.01$$

$$p_8 = \frac{128}{40320} = 0.00317 < 0.01$$

Smallest value of n is 8.

6. A lamp is controlled by a worn push-button switch. Each time the switch is pressed, the lamp changes state (from on to off, or from off to on) with probability p, where $0 < p < 1$, and otherwise stays as it is, independently of what happens at any other press. The lamp is on before the switch is first pressed, and the probability that the lamp is on after the switch has been pressed n times is denoted by L_n.

(a) Show that $L_n = \frac{1}{2}[1 + (1-2p)^n]$. [3]

(b) Without using the result in part (a), explain why $L_n = \frac{1}{2}$ for every positive integer n when $p = \frac{1}{2}$. [1]

(c) For a particular switch, $p = 0.9$. Find L_{10}. [1]

(d) For a different switch, $L_2 = 0.58$. Find the possible values of p. [2]

(e) For any switch, explain, in context, what happens to L_n as n becomes large. [1]

Solution:

(a) Let X be the number of the n presses that change the state of the lamp. The n presses are independent and each changes the state with the same probability p, so $X \sim \mathrm{B}(n, p)$. Every change of state switches the lamp between on and off, and the lamp starts on, so after n presses the lamp is on exactly when it has changed state an even number of times, that is, when X is even. [B1]

Hence, with $q = 1 - p$, $L_n = \mathrm{P}(X \text{ even}) = \sum_{k \text{ even}} \binom{n}{k} p^k q^{n-k}$.

By the binomial theorem,

$(q + p)^n = \sum_{k=0}^{n} \binom{n}{k} p^k q^{n-k} = \mathrm{P}(X \text{ even}) + \mathrm{P}(X \text{ odd})$,

$(q - p)^n = \sum_{k=0}^{n} \binom{n}{k} (-p)^k q^{n-k} = \mathrm{P}(X \text{ even}) - \mathrm{P}(X \text{ odd})$, since $(-p)^k = p^k$ when k is even and $(-p)^k = -p^k$ when k is odd. [M1]

Adding, with $q + p = 1$ and $q - p = 1 - 2p$: $1 + (1 - 2p)^n = 2\,\mathrm{P}(X \text{ even})$, so $L_n = \frac{1}{2}[1 + (1 - 2p)^n]$. [A1]

Alternative route (full marks): conditioning on the state just before the $(n + 1)$th press, $L_{n+1} = (1 - p)L_n + p(1 - L_n) = p + (1 - 2p)L_n$ [B1]. Then $L_{n+1} - \frac{1}{2} = (1 - 2p)\left(L_n - \frac{1}{2}\right)$, so $L_n - \frac{1}{2}$ is a geometric sequence with common ratio $1 - 2p$ [M1]; since $L_1 = 1 - p$ gives $L_1 - \frac{1}{2} = \frac{1}{2}(1 - 2p)$, $L_n - \frac{1}{2} = \frac{1}{2}(1 - 2p)^n$ and $L_n = \frac{1}{2}[1 + (1 - 2p)^n]$ [A1].

✓ **Check:** $L_1 = 1 - p = \frac{1}{2}[1 + (1 - 2p)]$ and $L_2 = (1 - p)^2 + p^2 = 1 - 2p + 2p^2 = \frac{1}{2}[1 + (1 - 2p)^2]$.

Verifying the formula for particular values of n only does not show it.

(b) Consider the last (nth) press. Whatever the state of the lamp just before it, this press changes the state with probability $\frac{1}{2}$ and leaves it unchanged with probability $\frac{1}{2}$, so $L_n = \frac{1}{2}L_{n-1} + \frac{1}{2}(1 - L_{n-1}) = \frac{1}{2}$ for every $n \geq 1$: the last press alone decides whether the lamp is on, and the earlier presses make no difference. [B1]

(Consistent with part (a): $p = \frac{1}{2}$ makes $(1 - 2p)^n = 0$.)

(c) With $p = 0.9$, $1 - 2p = -0.8$, so $L_{10} = \frac{1}{2}[1 + (-0.8)^{10}] = \frac{1}{2}(1 + 0.107374) = 0.553687 = 0.554$ (3 s.f.). [B1]

(d) $L_2 = \frac{1}{2}[1 + (1 - 2p)^2] = 0.58$, so $(1 - 2p)^2 = 0.16$ and $1 - 2p = 0.4$ or $1 - 2p = -0.4$. [M1: the formula at $n = 2$ solved for $(1 - 2p)^2$, both square roots taken]

So $p = 0.3$ or $p = 0.7$. Both values lie in $0 < p < 1$, so both are possible: a switch that changes the state with probability 0.3 and one that changes it with probability 0.7 give the same L_2 (check: $0.7^2 + 0.3^2 = 0.58$ either way), because after an even number of presses the two switches cannot be told apart. [A1: both values; rejecting one of them earns A0]

(e) For $0 < p < 1$, $-1 < 1 - 2p < 1$, so $(1 - 2p)^n \to 0$ and $L_n \to \frac{1}{2}$ as $n \to \infty$. In context: after a large number of presses the lamp is almost equally likely to be on or off, whatever the value of p — the fact that it started on no longer affects its state. [B1] (For $p = 0.9$ the values of L_n lie alternately above and below $\frac{1}{2}$ and close in on $\frac{1}{2}$; for $p < \frac{1}{2}$ they decrease towards $\frac{1}{2}$ from above.)

7. A spinner can land on any one of the scores 1, 2, 3, 4 and 5, and it is fair if these five scores are equally likely. Wei suspects that his spinner favours high scores. He spins it 50 times and records only T, the total of the 50

scores, and he assumes that the variance of the score on a single spin of his spinner is the same as for a fair spinner.

(a) State hypotheses for a test of Wei's suspicion, defining any parameters that you use. [1]

(b) Find the least value of T that would lead Wei to conclude, at the 5% level of significance, that his spinner favours high scores. [4]

(c) Explain why Wei's test is valid although the score on a single spin is not normally distributed. [1]

(d) Nadia spins a different spinner 50 times and obtains a total of 150. She says that this shows that her spinner is fair. Explain why Nadia is wrong, and give the probabilities of the five scores for a spinner that is not fair but whose score has the same mean and the same variance as the score on a fair spinner. [2]

Solution:

(a) Let μ be the population mean score on one spin of Wei's spinner. $H_0: \mu = 3$ (the spinner is fair), $H_1: \mu > 3$ (the spinner favours high scores). [B1]

(b) Let X be the score on one spin. Under H_0 the spinner is fair, so $P(X = x) = \frac{1}{5}$ for $x = 1,2,3,4,5$.

$E(X) = \frac{1+2+3+4+5}{5} = 3$, $E(X^2) = \frac{1+4+9+16+25}{5} = 11$, so $Var(X) = 11 - 3^2 = 2$. [B1]

The 50 scores are independent, so under H_0, $E(T) = 50 \times 3 = 150$ and $Var(T) = 50 \times 2 = 100$ (not $50^2 \times 2$). Since $n = 50$ is large, by the Central Limit Theorem $T \sim N(150,100)$ approximately. [M1]

Wei concludes that the spinner favours high scores when $P(T \geq t) \leq 0.05$ under H_0, i.e. when $\frac{t-150}{10} \geq 1.6449$, so $t \geq 150 + 1.6449 \times 10 = 166.45$. [M1]

(Equivalently, with the mean score $\bar{X} \sim N\left(3, \frac{2}{50}\right)$ approximately, $\bar{x} > 3 + 1.6449 \times 0.2 = 3.329$ and $T = 50\bar{x} > 166.45$.)

T is an integer, so the least value of T is 167. [A1]

✓ **Check:** $P(T \geq 167) = P(Z \geq 1.7) = 0.0446 < 0.05$, while $P(T \geq 166) = P(Z \geq 1.6) = 0.0548 > 0.05$.

(c) The score on a spin has a discrete uniform distribution, not a normal one, but the sample of 50 spins is large ($n \geq 30$), so by the Central Limit Theorem the total T of the 50 independent scores (equivalently their mean) is approximately normally distributed, so the z-test in (b) is valid. [B1]

(d) A total of 150 is exactly the expected total under H_0, so the test would not reject H_0; but not rejecting H_0 only means there is insufficient evidence that the spinner is unfair — it does not show that it is fair. The total depends on the individual scores only through their mean and variance: any spinner whose mean score is 3 has expected total 150, and a spinner whose score has mean 3 and variance 2 gives $T \sim N(150,100)$ approximately, exactly as a fair spinner does, so no value of T can tell the two apart. [B1]

Example: take $P(1) = P(5) = a$ and $P(3) = 1 - 2a$; the mean is 3 by symmetry and the variance is $a(1-3)^2 + a(5-3)^2 = 8a = 2$, so $a = \frac{1}{4}$: $P(1) = \frac{1}{4}$, $P(2) = 0$, $P(3) = \frac{1}{2}$, $P(4) = 0$, $P(5) = \frac{1}{4}$. Check: mean $= \frac{1}{4} + \frac{3}{2} + \frac{5}{4} = 3$, $E(X^2) = \frac{1}{4} + \frac{9}{2} + \frac{25}{4} = 11$, variance $= 11 - 9 = 2$. [B1]

(Any non-uniform distribution on 1 to 5 with mean 3 and variance 2 earns the mark, e.g. $P(1) = P(5) = \frac{1}{6}$, $P(2) = P(4) = \frac{1}{3}$, $P(3) = 0$; the symmetric family is $P(1) = P(5) = a$, $P(2) = P(4) = 1 - 4a$, $P(3) = 6a - 1$ with $\frac{1}{6} \leq a \leq \frac{1}{4}$, $a \neq \frac{1}{5}$.)

8. A student hangs loads of x newtons from a spring and measures the length, y cm, of the spring for each load. From her five pairs of values she calculated the equation of the least squares regression line of y on x as $y = 1.6x + 12.3$, but she later lost two of the lengths, shown as p and q in the table.

Load, x (N)	2	4	6	8	10
Length, y (cm)	15.7	p	21.8	q	28.6

(a) Find the values of p and q. [4]

(b) Use a suitable regression line to estimate the load for which the length of the spring is 20.0 cm, explaining your choice of line. Explain whether your estimate is reliable. [3]

(c) The student then repeats the measurement for a load of 6 N and records a length of 21.9 cm. Without doing any detailed calculation, explain whether including this sixth pair of values with the other five would change

(i) the equation of the regression line of y on x, [2]

(ii) the value of the product moment correlation coefficient. [1]

Solution:

(a) The least squares regression line of y on x passes through the mean point $(\bar{x}, \bar{y})$.

$\bar{x} = \frac{2+4+6+8+10}{5} = 6$, so $\bar{y} = 1.6(6) + 12.3 = 21.9$. [M1]

$\sum y = 5 \times 21.9 = 109.5$, so $15.7 + p + 21.8 + q + 28.6 = 109.5$, giving $p + q = 43.4$. [A1]

The gradient of the line is $b = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$ (MF26). The values of $x - \bar{x}$ are $-4, -2, 0, 2, 4$, so $\sum(x - \bar{x})^2 = 16 + 4 + 0 + 4 + 16 = 40$.

Since $\sum(x - \bar{x}) = 0$, $\sum(x - \bar{x})(y - \bar{y}) = \sum(x - \bar{x})y = -4(15.7) - 2p + 0(21.8) + 2q + 4(28.6) = 51.6 + 2(q - p)$. [M1]

$b = 1.6$: $\frac{51.6+2(q-p)}{40} = 1.6$, so $51.6 + 2(q - p) = 64$ and $q - p = 6.2$.

With $p + q = 43.4$: $2q = 49.6$, so $q = 24.8$ and $p = 18.6$. [A1]

(Alternative for the gradient: $\sum xy = 2(15.7) + 4p + 6(21.8) + 8q + 10(28.6) = 448.2 + 4p + 8q$ and $\sum x = 30$, so $\sum xy - \frac{\sum x \sum y}{5} = 448.2 + 4p + 8q - \frac{30 \times 109.5}{5} = 4p + 8q - 208.8$; $\sum x^2 - \frac{(\sum x)^2}{5} = 220 - 180 = 40$; $\frac{4p+8q-208.8}{40} = 1.6$ gives $p + 2q = 68.2$, and with $p + q = 43.4$, $q = 24.8$, $p = 18.6$.)

✓ **Check:** with $y = 15.7, 18.6, 21.8, 24.8, 28.6$ the calculator gives the line $y = 1.6x + 12.3$; the residuals are $0.2, -0.1, -0.1, -0.3, 0.3$.

(b) The loads x were chosen by the student and the lengths y were measured, so x is the controlled (independent) variable and y the dependent one. The regression line of y on x is therefore the suitable line, even though it is the load that is being estimated; the line of x on y is not used. [B1]

$20.0 = 1.6x + 12.3 \Rightarrow x = \frac{7.7}{1.6} = 4.8125$, so the estimated load is 4.81 N (3 s.f.). [B1]

(The line of x on y, $x = 0.624y - 7.66$, would give 4.82 N: this earns neither mark.)

With the completed data, $r = 0.99883 = 0.999$ (3 s.f.), which is very close to 1, so there is a very strong positive linear correlation between x and y; and 20.0 cm lies within the range of the observed lengths, 15.7 cm to 28.6 cm (the estimated load lies within 2 N to 10 N), so the estimate is an interpolation. The estimate is reliable. [B1] (both reasons needed)

(c)(i) From (a), $\bar{x} = 6$ and $\bar{y} = 21.9$, so the new pair $(6, 21.9)$ is exactly the mean point $(\bar{x}, \bar{y})$ of the five pairs (it lies on the regression line, at its mean point). With it included, the means of the six values of x and of y are $\frac{30+6}{6} = 6$ and $\frac{109.5+21.9}{6} = 21.9$: unchanged. [B1]

For the new pair, $x - \bar{x} = 0$ and $y - \bar{y} = 0$, so it adds 0 to both $\sum(x - \bar{x})(y - \bar{y})$ and $\sum(x - \bar{x})^2$; the gradient b is still $\frac{64}{40} = 1.6$ and the line still passes through $(6, 21.9)$. So the equation of the regression line of y on x does not change: it is still $y = 1.6x + 12.3$. [B1]

(Also accepted: the new point lies on the line $y = 1.6x + 12.3$, so its residual for this line is 0. For any line, the sum of the squared residuals of the six points is at least the sum for the original five points, which is least for $y = 1.6x + 12.3$; and for $y = 1.6x + 12.3$ the sixth point adds 0. So $y = 1.6x + 12.3$ still minimises the sum of the squared residuals.)

(c)(ii) $r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}}$. The means are unchanged and both deviations of the new point are 0, so all three sums, 64, 40 and 102.64, are unchanged, and r does not change: it is still 0.999. [B1]

(Adding a point that lies exactly on the line does NOT strengthen the correlation here: the answer 'r increases' earns B0.)

9. A bag contains 1 red ball and 1 blue ball. A ball is taken at random from the bag and its colour is noted; the ball is then returned to the bag together with one more ball of the same colour. This process is repeated.

(a) Find the probability that the first two balls taken are the same colour. [2]

(b) The colours of the first three balls taken are written in order; for example, RBB means that the first ball taken is red and the second and third are blue. Find the probabilities of RRB, RBR and BRR, and explain why these three probabilities are equal. [3]

(c) Given that the third ball taken is red, find the probability that the first ball taken is red. Explain why your answer is not equal to $\frac{1}{2}$, the probability that the first ball taken is red. [3]

(d) The random variable X is the number of red balls among the first n balls taken. Find the probability distribution of X. [3]

(e) Find the probability that at least 90 of the first 99 balls taken are the same colour. Explain why this probability is much greater than it would be if each ball taken were returned to the bag with no extra ball added. [2]

Solution:

(a) After a red ball is taken the bag holds 2 red and 1 blue; after a blue, 1 red and 2 blue.

$\mathrm{P(RR)} = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$, $\mathrm{P(BB)} = \frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$

P(first two the same colour) $= \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$.

(b) $\mathrm{P(RRB)} = \frac{1}{2} \times \frac{2}{3} \times \frac{1}{4} = \frac{1}{12}$ (after RR the bag holds 3 red, 1 blue)

$\mathrm{P(RBR)} = \frac{1}{2} \times \frac{1}{3} \times \frac{2}{4} = \frac{1}{12}$ (after RB the bag holds 2 red, 2 blue)

$\mathrm{P(BRR)} = \frac{1}{2} \times \frac{1}{3} \times \frac{2}{4} = \frac{1}{12}$ (after BR the bag holds 2 red, 2 blue)

Why they are equal: exactly one ball is added after every draw, whatever its colour, so the bag holds 2, 3 and 4 balls at the first, second and third draws; the denominators are 2,3,4 in every order. A red ball is added only when a red ball is taken, so the first red ball taken is taken when the bag holds 1 red ball and the second when it holds 2 red balls, wherever they come in the sequence; likewise the one blue ball is taken when the bag holds 1 blue ball. So the numerators are 1,2 (for the reds) and 1 (for the blue) in some order, and each probability is $\frac{1\times2\times1}{2\times3\times4} = \frac{1}{12}$.

(c) The sequences with a red third ball are RRR, RBR, BRR and BBR.

$\mathrm{P(RRR)} = \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = \frac{1}{4}$, $\mathrm{P(BBR)} = \frac{1}{2} \times \frac{2}{3} \times \frac{1}{4} = \frac{1}{12}$, and $\mathrm{P(RBR)} = \mathrm{P(BRR)} = \frac{1}{12}$ from (b).

P(third red) $= \frac{1}{4} + \frac{1}{12} + \frac{1}{12} + \frac{1}{12} = \frac{1}{2}$

[or: the rule treats the two colours alike and the bag starts with one ball of each, so every draw is equally likely to be red or blue].

P(first red and third red) $= \mathrm{P(RRR)} + \mathrm{P(RBR)} = \frac{1}{4} + \frac{1}{12} = \frac{1}{3}$

P(first red | third red) $= \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}$.

[Using (b): $\mathrm{P(RBR)} = \mathrm{P(RRB)}$, so P(first red and third red) $= \mathrm{P(RRR)} + \mathrm{P(RRB)} =$ P(first two red) $= \frac{1}{3}$ from (a).]

Why it is not $\frac{1}{2}$: the colour of the first ball changes the contents of the bag for the later draws. If the first ball is red, an extra red ball is added, so a red third ball is more likely than if the first ball is blue; the events 'first ball red' and 'third ball red' are therefore not independent, and knowing that the third ball is red makes it more likely that the first ball was red.

(d) Take any one sequence of n colours containing k reds and $n - k$ blues, where $0 \le k \le n$.

Before the rth draw, $r - 1$ balls have been added, so the bag holds $r + 1$ balls whatever colours were taken: the denominators multiply to $2 \times 3 \times \cdots \times (n + 1) = (n + 1)!$.

A red ball is added only when a red ball is taken, so the jth red ball taken is taken from a bag holding j red balls: the red draws give the numerators $1 \times 2 \times \cdots \times k = k!$. In the same way the blue draws give $(n - k)!$.

So every sequence with k reds has probability $\frac{k!\,(n-k)!}{(n+1)!}$, which depends only on k.

There are ${}^{n}C_{k} = \frac{n!}{k!\,(n-k)!}$ such sequences, so

$\mathrm{P}(X = k) = \frac{n!}{k!\,(n-k)!} \times \frac{k!\,(n-k)!}{(n+1)!} = \frac{n!}{(n+1)!} = \frac{1}{n+1}$, for $k = 0,1,2,\ldots,n$.

X is equally likely to take each of the $n + 1$ values $0,1,\ldots,n$. [Check, $n = 3$: BBB and RRR each $\frac{1}{4}$; the three sequences with one red and the three with two reds each $\frac{1}{12}$, so each value of X has probability $\frac{1}{4}$.]

(e) With $n = 99$, $\mathrm{P}(X = k) = \frac{1}{100}$ for each $k = 0,1,\ldots,99$.

At least 90 the same colour means $X \geq 90$ (at least 90 red) or $X \leq 9$ (at least 90 blue); these cannot both happen, since $90 + 90 > 99$.

$X \geq 90$: the 10 values $90,\ldots,99$; $X \leq 9$: the 10 values $0,\ldots,9$.

Probability $= \frac{20}{100} = \frac{1}{5}$.

Explanation: because each ball taken goes back with another of its colour, a colour that is taken becomes more likely to be taken again (as in (a), the first two balls match with probability $\frac{2}{3}$, not $\frac{1}{2}$). Whichever colour happens to be taken more often in the early draws therefore tends to be taken more often afterwards: an early imbalance is reinforced, not corrected, and the number of reds is spread evenly over 0 to 99. If each ball were returned with no extra ball, every draw would be red with probability $\frac{1}{2}$ independently of the others, so the number of reds would be $\mathrm{B}(99, \frac{1}{2})$, with mean 49.5 and standard deviation about 5, and 90 or more of one colour would be almost impossible (probability about 6×10^{-18}).

10. In this question you should state the parameters of any distributions you use.

A factory uses a gauge to check metal rods that are intended to be 200 mm long. Each rod is measured 3 times and is rejected if its recorded length is less than 199.2 mm or greater than 200.8 mm; under the mean rule the recorded length is the mean of the 3 readings, and under the median rule it is the median of the 3 readings. In parts (a) to (d), the 3 readings of a rod whose length is exactly 200 mm are modelled as independent observations of a random variable with the distribution $\mathrm{N}(200, 0.5^2)$.

(a) Find the probability that a rod of length exactly 200 mm is rejected under the mean rule. [2]

(b) Find the probability that a rod of length exactly 200 mm is rejected under the median rule, explaining your reasoning. [4]

(c) Suggest a reason why the factory might prefer the median rule, in spite of your answers to parts (a) and (b). [1]

(d) Find the value of t such that, when the limits for the median rule are $(200 - t)$ mm and $(200 + t)$ mm, the probability that a rod of length exactly 200 mm is rejected under the median rule is 0.01. [3]

(e) In fact the lengths of the rods vary. In a refined model, the length, in mm, of a randomly chosen rod has the distribution $\mathrm{N}(200, 0.3^2)$, and each reading of the rod is equal to its length plus an error, in mm, which has the distribution $\mathrm{N}(0, 0.4^2)$; the 3 errors are independent of one another and of the length of the rod.

(i) Explain why the 3 readings of a randomly chosen rod are not independent. [1]

(ii) Find the probability that a randomly chosen rod is rejected under the mean rule. [3]

Solution:

(a) Let X mm be a reading of a rod of length exactly 200 mm, so $X \sim \mathrm{N}(200, 0.5^2)$. The mean of 3 independent readings is $\bar{X} \sim \mathrm{N}\left(200, \frac{0.5^2}{3}\right) = \mathrm{N}\left(200, \frac{1}{12}\right)$. [B1]

$\mathrm{P}(\text{rejected under the mean rule}) = \mathrm{P}(\bar{X} < 199.2) + \mathrm{P}(\bar{X} > 200.8) = 1 - \mathrm{P}(199.2 \leq \bar{X} \leq 200.8) = 2 \times 0.0027918 = 0.00558$. [A1]

(b) Place the 3 readings in order $R_1 \le R_2 \le R_3$; the median is R_2. If $R_2 > 200.8$, then $R_3 \ge R_2 > 200.8$ as well, so at least two readings exceed 200.8; conversely, if at least two readings exceed 200.8, then the two largest, R_2 and R_3, both do, so $R_2 > 200.8$. Hence the median exceeds 200.8 exactly when at least 2 of the 3 readings exceed 200.8, and likewise the median is less than 199.2 exactly when at least 2 readings are less than 199.2. [B1]

$q = \mathrm{P}(X > 200.8) = 0.054799$. [B1]

Let Y be the number of the 3 readings that exceed 200.8. The readings are independent and each exceeds 200.8 with the same probability q, so $Y \sim \mathrm{B}(3, 0.054799)$ and

$\mathrm{P}(\text{median} > 200.8) = \mathrm{P}(Y \ge 2) = 3q^2(1-q) + q^3 = 0.0086798$. [M1]

By symmetry $\mathrm{P}(\text{median} < 199.2) = 0.0086798$ also, and the two events cannot both happen (with only 3 readings, at least 2 cannot lie above 200.8 while at least 2 lie below 199.2), so

$\mathrm{P}(\text{rejected under the median rule}) = 2 \times 0.0086798 = 0.0174$. [A1]

(Using one tail only gives 0.00868; using $\mathrm{P}(Y \ge 1)$ confuses the median with the largest reading.)

(c) If one of the 3 readings is wildly wrong (for example, because of dirt on the rod or a slip of the gauge), the mean is pulled towards that reading, so a good rod may be rejected (or a faulty rod accepted); the median is the middle reading and is not affected by a single extreme reading. [B1]

(d) Let $p = \mathrm{P}(X > 200 + t) = \mathrm{P}(X < 200 - t)$. As in part (b), the rod is rejected under the median rule when at least 2 readings exceed $200 + t$ or at least 2 readings are less than $200 - t$, so $2[3p^2(1-p) + p^3] = 0.01$, that is, $3p^2 - 2p^3 = 0.005$. [M1]

From the GC, the roots of $2p^3 - 3p^2 + 0.005 = 0$ are -0.0403, 0.041400 and 1.50. Since p is a probability and $t > 0$ requires $p < 0.5$, $p = 0.041400$. [M1]

Then $\mathrm{P}(X > 200 + t) = 0.041400$ gives $200 + t = 200.867$ (that is, $\frac{t}{0.5} = 1.7347$), so $t = 0.867$. [A1]

✓ **Check:** $\mathrm{P}(X > 200.867) = 0.0414$ and $2[3(0.0414)^2 - 2(0.0414)^3] = 0.0100$.

(e)(i) Each reading of a rod is its length L plus an error, so all 3 readings of the same rod contain the same value of L: a rod longer than 200 mm tends to give 3 readings above 200 mm, so the value of one reading affects the probabilities for the others and the readings are not independent. [B1]

(Each single reading $L + E$ still has the distribution $\mathrm{N}(200, 0.3^2 + 0.4^2) = \mathrm{N}(200, 0.5^2)$, exactly as in parts (a) to (d); only the independence has changed.)

(e)(ii) Let L mm be the length of the rod and E_1, E_2, E_3 mm the three errors, where $L \sim \mathrm{N}(200, 0.3^2)$, $E_i \sim \mathrm{N}(0, 0.4^2)$, all independent. The readings are $L + E_1$, $L + E_2$, $L + E_3$, so the mean of the 3 readings is $M = L + \frac{E_1 + E_2 + E_3}{3}$ (the length is not averaged away). [M1]

$\mathrm{E}(M) = 200$ and $\mathrm{Var}(M) = 0.3^2 + \frac{3 \times 0.4^2}{9} = 0.09 + \frac{0.16}{3} = \frac{43}{300} = 0.14333$, so $M \sim \mathrm{N}\left(200, \frac{43}{300}\right)$. [A1]

$\mathrm{P}(\text{rejected}) = 1 - \mathrm{P}(199.2 \le M \le 200.8) = 2 \times 0.017297 = 0.0346$. [A1]

(Treating the 3 readings as independent $\mathrm{N}(200, 0.5^2)$ observations gives variance $\frac{0.25}{3}$ and 0.00558 again, the answer to part (a): averaging reduces only the error part of the variance, not the variation between rods, so the rejection rate is about 6 times as large.)

Answers

1 (a) $x > -\frac{1}{2}$; (b) $\frac{x+2}{3}$; (c) $-\frac{1}{2} < x < 0$ or $x > \sqrt{7} - 1$

2 (a) $\left(\mathrm{e}, \frac{1}{\mathrm{e}}\right)$, a maximum point; (b) shown; (c) $3^{\pi} > \pi^3$; (d) $n = 2,3,4$

3 (a) $\mathrm{f}^{-1}(x) = \sqrt{1-x}$, domain $x \leq 1$; (b) $y = \mathrm{f}(x)$ from the end-point $(0,1)$ decreasing through $(1,0)$, $y = \mathrm{f}^{-1}(x)$ its reflection in $y = x$ from the end-point $(1,0)$ through $(0,1)$; (c) $(0,1)$, $(1,0)$ and $\left(\frac{\sqrt{5}-1}{2}, \frac{\sqrt{5}-1}{2}\right)$, and no others; (d) $\mathrm{g}(a) = b$ and $\mathrm{g}(b) = a$ with $a \neq b$ contradicts g increasing, so $a = b$; (e) $\left(-\frac{1+\sqrt{5}}{2}, -\frac{1+\sqrt{5}}{2}\right)$

4 (a) Depth increasing at $\frac{4}{3}$ cm per second; radius of the water surface changing at $-\frac{4}{3}$ cm per second (decreasing). (b) $\frac{19\pi}{12}$ s. (c) π cm (half the capacity, and the vase is symmetric about its narrowest section $y = \pi$). (d) Greatest 3 cm per second at $h = \pi$; least $\frac{3}{4}$ cm per second at $h = 0$ and $h = 2\pi$.

5 (a) 6; (b) 16 (shown); (c) 1 is the smallest integer, so it is first if it is on the increasing side and last if it is on the decreasing side; (d) $n = 8$

6 (a) $L_n = \frac{1}{2}[1 + (1-2p)^n]$ (shown). (b) The last press alone decides: whatever the state before it, it changes the state with probability $\frac{1}{2}$, so the lamp is on after it with probability $\frac{1}{2}$. (c)(i) 0.554. (c)(ii) $n = 1,3,5, \ldots, 17$. (d) $L_n \to \frac{1}{2}$ because $|1-2p| < 1$: after many presses the lamp is equally likely to be on or off, whatever p is, and its starting state no longer matters.

7 (a) $\mathrm{H}_0: \mu = 3$, $\mathrm{H}_1: \mu > 3$, μ the population mean score on one spin. (b) 167. (c) $n = 50$ is large, so by the Central Limit Theorem T is approximately normal. (d) Not rejecting H_0 does not prove fairness; the total reflects only the mean (and variance) of a score. E.g. $\mathrm{P}(1) = \mathrm{P}(5) = \frac{1}{4}$, $\mathrm{P}(3) = \frac{1}{2}$, $\mathrm{P}(2) = \mathrm{P}(4) = 0$.

8 (a) $p = 18.6$, $q = 24.8$. (b) Line of y on x (the load x is controlled by the student): 4.81 N; reliable, since $r = 0.999$ is close to 1 and 20.0 cm is within the data (interpolation). (c)(i) No change: $(6, 21.9)$ is $(\bar{x}, \bar{y})$, so the means and the gradient are unchanged; still $y = 1.6x + 12.3$. (ii) No change: every sum in r is unchanged, so $r = 0.999$ still.

9 (a) $\frac{2}{3}$; (b) each $\frac{1}{12}$: the bag holds 2,3,4 balls at the three draws whatever the colours, and the reds are taken from 1 then 2 red balls and the blue from 1 blue ball in any order; (c) $\frac{2}{3}$: a red first ball adds a red ball, making a red third ball more likely, so the events are not independent; (d) $\mathrm{P}(X = k) = \frac{1}{n+1}$, $k = 0,1, \ldots, n$; (e) $\frac{1}{5}$: a colour once taken is more likely to be taken again, so an early lead is reinforced, whereas with plain replacement the number of reds is $\mathrm{B}(99, \frac{1}{2})$ and such an imbalance is almost impossible

10 (a) 0.00558. (b) 0.0174. (c) The median is not affected by one wildly wrong reading, which would pull the mean and could reject a good rod. (d) $t = 0.867$. (e)(i) All 3 readings of a rod contain the same length L of that rod, so they are not independent. (e)(ii) 0.0346.